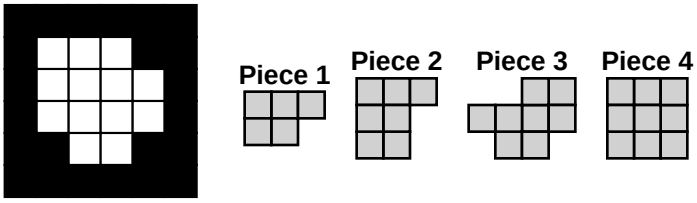
Math Kangaroo Practice Exam - Grades 3-4

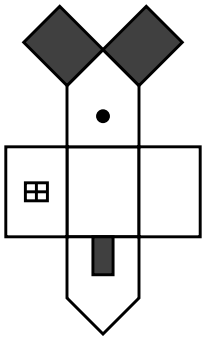
20 Randomly Selected Practice Questions (Similar to Contest Problems — Never Originals)

Question 1. The picture shows a 6 × 6 square with some cells black and the rest white. We want to cover the white area completely without overlaps using two of the numbered pieces (1, 2, 3, 4). Which two pieces should we choose?

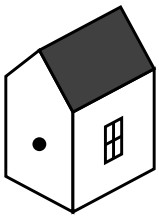


A) 1 and 2 **B)** 2 and 3 **C)** 1 and 3 **D)** 2 and 4 **E)** 3 and 4

Question 2. This picture is cut out and folded to make a house. Which house does it become? *(The dark parts at the top are the roof and the middle square is the floor; fold the walls away from you so the drawings are on the outside of the house.)*

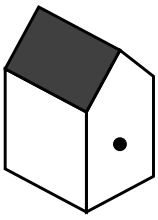


A)



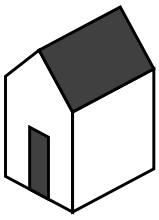
B)

C)



D)

E)



Question 3. Leo built a tunnel 3 cubes deep with a 5 × 5 outer frame and a 3 × 3 tunnel opening (fig.1). He then used some of those cubes to build a 3-layer stepped pyramid with 5 × 5, 3 × 3, and 1 × 1 layers (fig.2). How many cubes from the tunnel were left over?

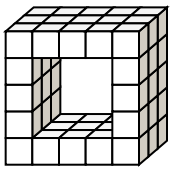


fig.1

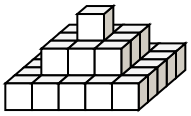


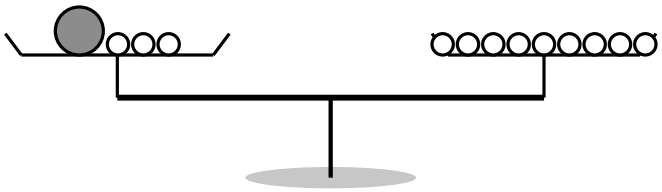
fig.2

A) 13 **B)** 15 **C)** 18 **D)** 21 **E)** 23

Question 4. What is the first number greater than **4090** such that the sum of its digits equals the sum of the digits of 4090?

A) 4099 **B)** 4100 **C)** 4190 **D)** 4180 **E)** 4108

Question 5. The balance is level. On the left plate there is 1 melon (big circle) and 3 oranges; on the right plate there are 9 oranges. How many oranges weigh the same as **2 melons**?

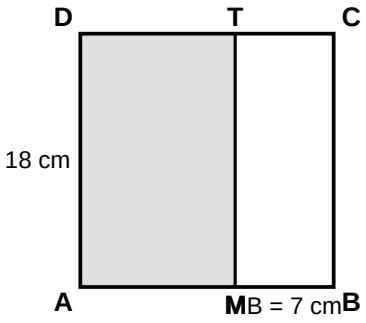


A) 6 **B)** 8 **C)** 9 **D)** 12 **E)** 18

Question 6. A bar code is made of 21 alternating black and white bars, starting and ending with a black bar. Each black bar is either wide or narrow. The number of white bars is 4 more than the number of wide black bars. How many narrow black bars are there?

A) 1 **B)** 2 **C)** 3 **D)** 4 **E)** 5

Question 7. The side of square *ABCD* is 18 cm. A line segment *MT* divides the square into two rectangles. The length of segment *MB* is 7 cm. How much greater is the perimeter of square *ABCD* than the perimeter of rectangle *AMTD*?

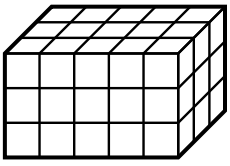


A) 12 cm **B)** 14 cm **C)** 16 cm **D)** 18 cm **E)** 22 cm

Question 8. How many different letters do the words **TRIANGLE** and **RECTANGLE** have in common?

A) 4 **B)** 5 **C)** 6 **D)** 7 **E)** 8

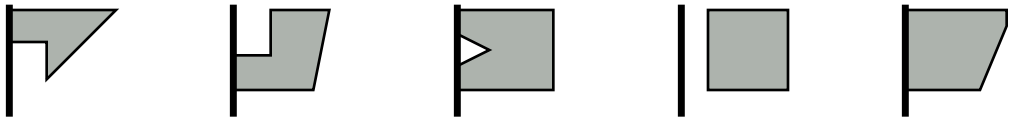
Question 9. Anna built this brick (5 cubes wide, 3 cubes tall, 3 cubes deep). The outside is completely red and the cubes inside are blue. How many **red** cubes did she use?



A) 30 **B)** 36 **C)** 39 **D)** 42 **E)** 45

Question 10. Four of the 5 flag silhouettes below are impossible unless the rectangular flag is torn from the pole. Which silhouette **can** happen with an intact flag?

A) **B)** **C)** **D)** **E)**



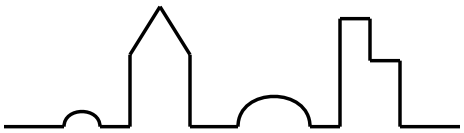
Question 11. Sam, Tara, Uma, and Vic each do a different sport: soccer, tennis, swimming, and cycling. Sam and Uma play sports with a ball, Tara is in the pool, and Sam does not play soccer. Which statement is **not** true?

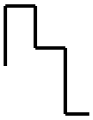
A) Sam plays tennis **B)** Uma plays tennis **C)** Tara does swimming
D) Vic does cycling **E)** Uma plays soccer

Question 12. Dan has cards 8, 2, and 1; Eva has cards 9, 7, and 8; and Finn has cards 5, 4, and 2. Using +, -, ×, ÷ and each of their own three cards once, who **cannot** obtain **24**?

A) Dan **B)** Eva **C)** Finn
D) All can get 24 **E)** Dan and Eva

Question 13. Here is a skyline with a small arch, a tower and a dome. Which piece **cannot** belong to it?

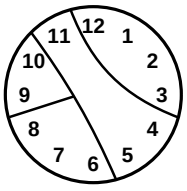
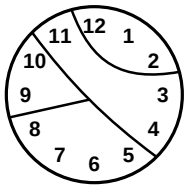
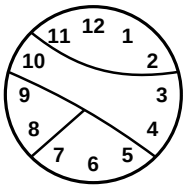
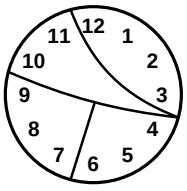
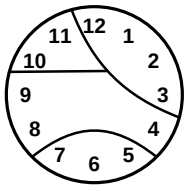


A) **B)** **C)**  **D)** **E)**

Question 14. Hoppy the frog's practice jump was 40 dm, 40 cm, and 40 mm long. In the final round, he jumped 135 cm farther than his practice jump. How long was Hoppy's final jump?

A) 4 m 44 cm **B)** 5 m 79 cm **C)** 5 m 39 cm
D) 6 m 15 cm **E)** 9 m 75 cm

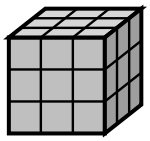
Question 15. The face of a clock is cracked into 4 pieces so that two of the pieces each have a sum of 18, and the other two pieces each have a sum of 21. Which clock face shows this?

A)  **B)**  **C)**  **D)**  **E)** 

Question 16. A locker hallway has 80 lockers numbered from **1 to 80**. If each digit is a plastic sticker, how many digit stickers are needed in total?

A) 151 **B)** 153 **C)** 160 **D)** 161 **E)** 169

Question 17. A 3 cm × 3 cm × 3 cm wooden cube is painted grey on the outside and cut into 27 small 1 cm cubes. How many of the small cubes have **exactly 1** face painted?

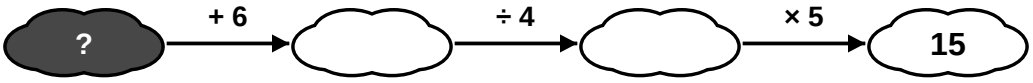


A) 4 **B)** 6 **C)** 8 **D)** 9 **E)** 12

Question 18. A bicycle trip counter shows the 4-digit number **1987**, whose digits are all different. After how many more kilometres will the counter next show a number with all four digits different?

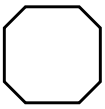
A) 3 **B)** 13 **C)** 16 **D)** 26 **E)** 36

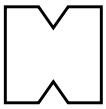
Question 19. Sofia wrote a number in the dark cloud and followed the sequence of calculations in the diagram to end up with **15**. What number did Sofia write in the dark cloud?

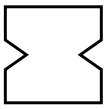


A) 6 **B)** 8 **C)** 10 **D)** 12 **E)** 14

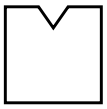
Question 20. A square piece of paper is folded in half vertically and then in half horizontally into a smaller square. One of its corners consists of **4 loose paper corners** (with no fold creases). If you cut off that corner and unfold the paper, which shape do you get?

A) 

B) 

C) 

D)

E) 

Answer Key - 20-Question Practice Exam

For generated/2026-10-19-19-33-30.pdf (Randomly sampled from question-pool.txt)

Solutions & kid-friendly explanations

Question 1 (inspired by 2003 #18): C The white area has 13 squares. Placing Piece 1 (5 squares) in the top-left of the white region leaves an 8-square stepped shape on the right and bottom that matches Piece 3 exactly.

Question 2 (inspired by 2003 #12): D Facing the door wall, the window wall is on your right and the plain wall on your left. Only D shows that.

Question 3 (inspired by 2002 #20): A Each of the 3 layers of the tunnel has 5×5 - 3×3 = 25 - 9 = 16 cubes, so the tunnel has 3 × 16 = 48 cubes. The pyramid uses 25 + 9 + 1 = 35 cubes. That leaves 48 - 35 = 13 cubes unused.

Question 4 (inspired by 2007 #4): E The sum of the digits of 4090 is 4 + 0 + 9 + 0 = 13. Every number from 4091 to 4099 has a digit sum larger than 13. Moving to the 4100s, we need the last two digits to sum to 13 - (4 + 1) = 8, which gives 4108.

Question 5 (inspired by 2002 #9): D Take 3 oranges off each side: 1 melon = 6 oranges, so 2 melons = 12 oranges.

Question 6 (inspired by 2003 #17): E Out of 21 alternating bars starting and ending with black, there are 11 black bars and 10 white bars. Since 10 white bars is 4 more than the wide black bars, there are 10 - 4 = 6 wide black bars, leaving 11 - 6 = 5 narrow black bars.

Question 7 (inspired by 2002 #6): B Since AB = 18 cm and MB = 7 cm, AM = 18 - 7 = 11 cm. Perimeter of square = 4 × 18 = 72 cm. Perimeter of rectangle AMTD = 2 × (18 + 11) = 58 cm. Difference = 72 - 58 = 14 cm (which is simply 2 × MB = 2 × 7 = 14 cm).

Question 8 (inspired by 2007 #3): D The distinct letters of TRIANGLE are {T, R, I, A, N, G, L, E} and of RECTANGLE are {R, E, C, T, A, N, G, L}. They share the 7 letters T, R, A, N, G, L, and E.

Question 9 (inspired by 2003 #9): D Total = 5 x 3 x 3 = 45. Inside (blue) = 3 x 1 x 1 = 3. Red = 45 - 3 = 42.

Question 10 (inspired by 2002 #15): E Only in E is the flag attached continuously along its full vertical segment on the flagpole while the free right edge folds slightly inward in the wind. In A, B, C, and D, parts of the flag are detached from the pole.

Question 11 (inspired by 2002 #18): B The two ball sports are soccer and tennis. Since Sam does not play soccer, Sam plays tennis and Uma plays soccer. Tara does swimming, leaving cycling for Vic. Thus, 'Uma plays tennis' (B) is not true.

Question 12 (inspired by 2002 #21): C Dan can get 24 with 8 × (2 + 1) = 24. Eva can get 24 with 9 + 7 + 8 = 24. Finn's cards (5, 4, 2) can make 5 × 4 + 2 = 22 or (5 + 2) × 4 = 28, but never 24. So Finn cannot obtain 24.

Question 13 (inspired by 2002 #7): D After each arch the line goes flat and then UP. Piece D goes down after the arch.

Question 14 (inspired by 2003 #10): B Convert the practice jump to centimetres: 40 dm = 400 cm, 40 cm = 40 cm, and 40 mm = 4 cm, giving 400 + 40 + 4 = 444 cm. Adding 135 cm gives 444 + 135 = 579 cm = 5 m 79 cm.

Question 15 (inspired by 2002 #11): E In clock E, the top-right piece has {12, 1, 2, 3} = 18, the bottom piece has {5, 6, 7} = 18, the top-left piece has {10, 11} = 21, and the middle piece has {8, 9, 4} = 21.

Question 16 (inspired by 2007 #23): A Lockers 1–9 need 9 × 1 = 9 stickers, and lockers 10–80 (71 lockers) need 71 × 2 = 142 stickers. That is 9 + 142 = 151 digit stickers in total.

Question 17 (inspired by 2007 #17): B A cube has 6 faces, and in a 3 × 3 × 3 cube only the single center cube on each face has 1 painted face. That gives 6 × 1 = 6 small cubes.

Question 18 (inspired by 2002 #24): D Numbers 1988–1999 all repeat 8 or 9, and numbers 2000–2012 repeat 0, 1, or 2. The first number after 1987 with four different digits is 2013 (digits 2, 0, 1, 3). That happens after 2013 - 1987 = 26 km.

Question 19 (inspired by 2007 #5): A Working backwards from 15: before × 5 the number was 15 ÷ 5 = 3; before ÷ 4 it was 3 × 4 = 12; and before + 6 it was 12 - 6 = 6.

Question 20 (inspired by 2007 #24): A The corner with 4 loose layers contains all 4 outer corners of the original square stacked together. Snipping it cuts off all 4 outer corners of the unfolded square, producing the octagon in A.